Artificial Intelligence Driven Meta-Analysis

Introduction to Meta Analysis



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Reminder for Group Project

1. **Date of Final Report：** Dec. 5, 2025
   - **Deadline of Submission：** 10.00 am
2. **Date of Presentation：** Dec. 12, 2025
   - **Time：** 10:00-13:00 and 19:00-22:00
   - **Deadline of Submission：** 10.00 am



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Major Requirements for Group Project

- Teams consist of **3 or 4 randomly** assigned members
- The project data will be released on **Sep 12, 2026** (today)
- Your team need to select **a topic** and utilize it to develop a detailed project.
- Teams are required to submit ***the final project reports with a Turnitin report, the ppt or pdf of the presentation slides, the work contribution and Gen AI documents***.
- Teams should select their **own presenter**.
- The project leader will submit **all project files**, and every member must submit a **peer evaluation for the entire team.**
- **For more details, please refer to the files in Canvas.**

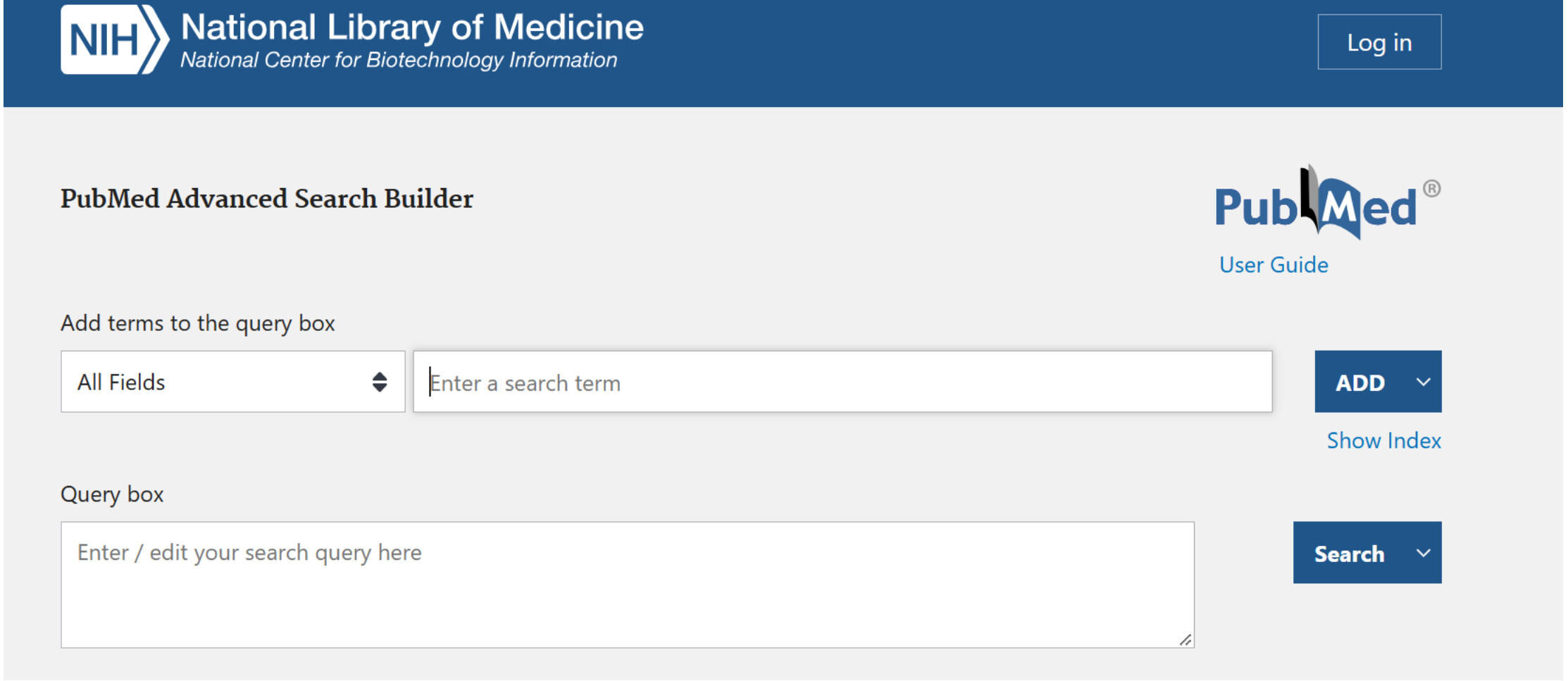


Systematic Review And Meta-Analysis

A Step by Step guide Approach


Garbage in
Garbage OUT

How to Search


National Library of Medicine
National Center for Biotechnology Information
Log in
PubMed Advanced Search Builder
PubMed
User Guide
Add terms to the query box
All Fields
Enter a search term
ADD
Show Index
Query box
Enter / edit your search query here
Search

Cochrane Library

Trusted evidence.
Informed decisions.
Better health.

Review language : English | Website language : English | Sign In

Cochrane reviews | Searching for trials | Clinical Answers | About | Help | About Cochrane

Advanced Search

Search | Search manager | Medical terms (MeSH) | PICO search

Save search | View saved searches | ? Search help

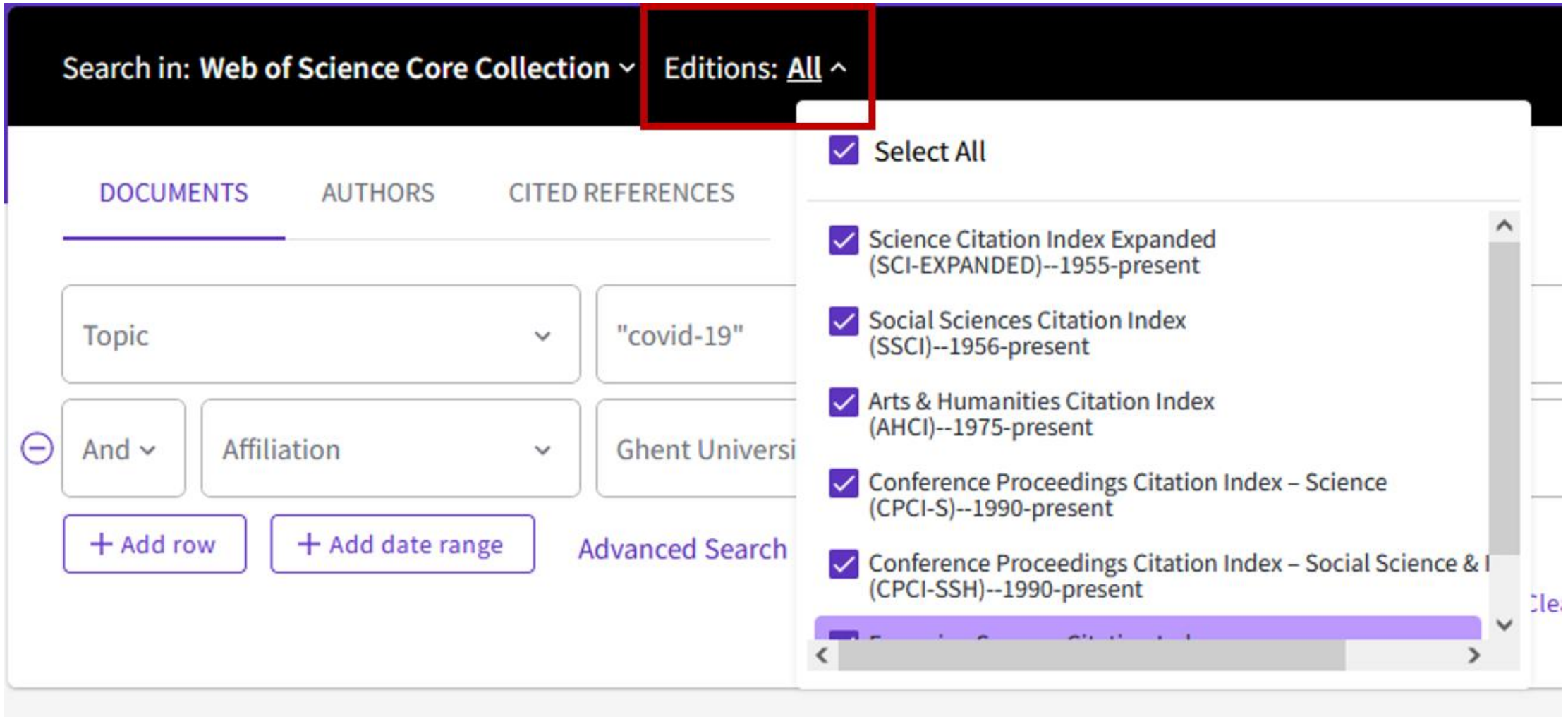
Did you know you can now select fields from Search manager using the S button (next to the search box)?
Search manager lets you add unlimited search lines, view results per line and access the MeSH browser using the new MeSH button.

− Title Abstract Keyword — Type a search term

− AND — Title Abstract Keyword — Type a search term

+

Search limits | Send to search manager | Run search


Search in: Web of Science Core Collection
Editions: All
DOCUMENTS
AUTHORS
CITED REFERENCES
Topic
"covid-19"
And
Affiliation
Ghent Universi
+ Add row
+ Add date range
Advanced Search
Select All
Science Citation Index Expanded (SCI-EXPANDED)--1955-present
Social Sciences Citation Index (SSCI)--1956-present
Arts & Humanities Citation Index (AHCI)--1975-present
Conference Proceedings Citation Index – Science (CPCI-S)--1990-present
Conference Proceedings Citation Index – Social Science & (CPCI-SSH)--1990-present

Embase

Search Emtree Journals Results My tools 4 Sign in

Quick PICO PV Wizard Medical device Advanced Drug Disease Device Citation information Query translator

Find articles by simple keyword search or build more complex queries

Search tips

Broad search

e.g. 'heart attack' AND stress

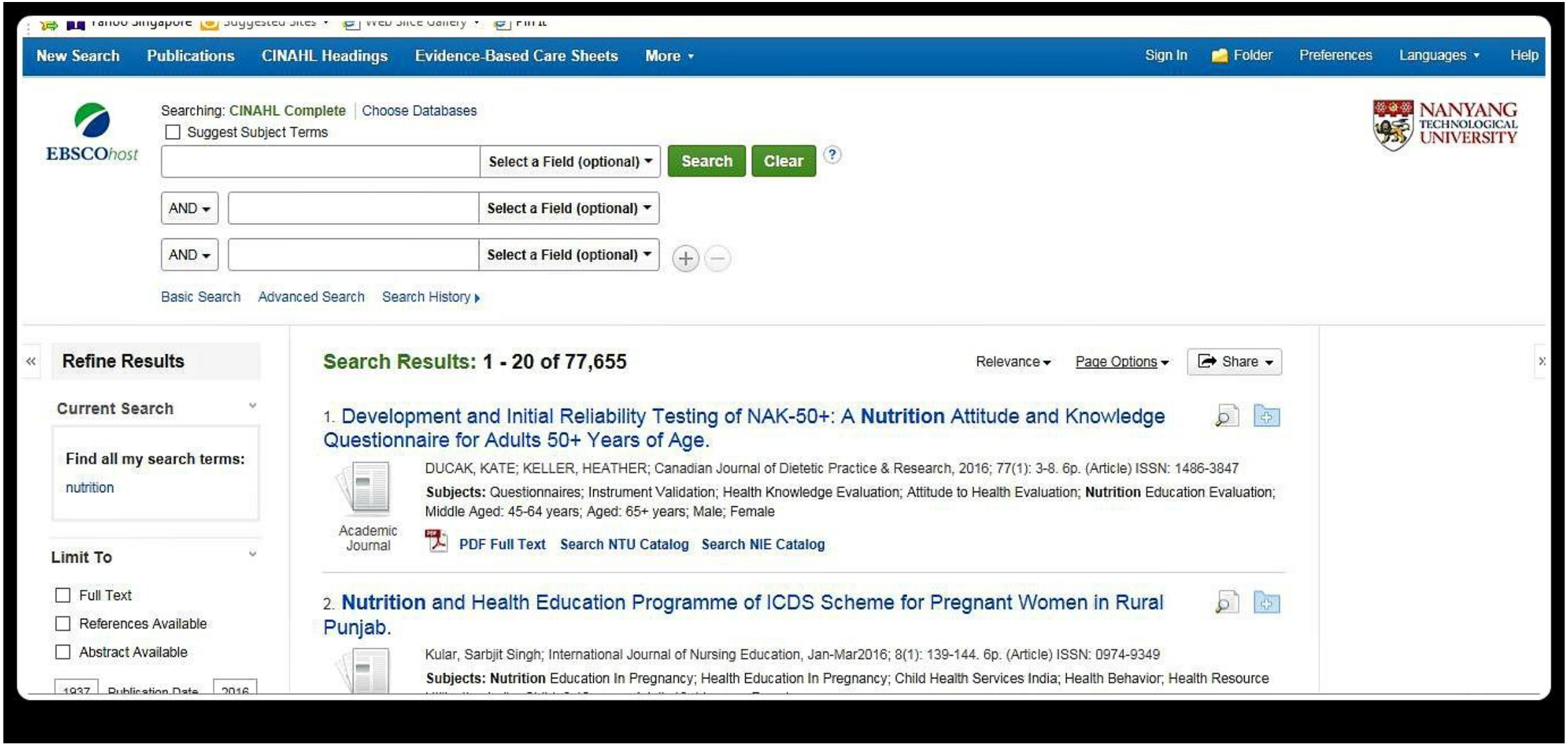
Title, Abstract or Author keywords

AND

e.g. zika virus

Add field Limit to Reset form

Show results


New Search
Publications
CINAHL Headings
Evidence-Based Care Sheets
More
Sign In
Folder
Preferences
Languages
Help
EBSCOhost
Searching: CINAHL Complete
Choose Databases
Suggest Subject Terms
Select a Field (optional)
Search
Clear
AND
Basic Search
Advanced Search
Search History
NANYANG TECHNOLOGICAL UNIVERSITY
Refine Results
Current Search
Find all my search terms:
nutrition
Limit To
Full Text
References Available
Abstract Available
Search Results: 1 - 20 of 77,655
Relevance
Page Options
Share
1. Development and Initial Reliability Testing of NAK-50+: A Nutrition Attitude and Knowledge Questionnaire for Adults 50+ Years of Age.
DUCAK, KATE; KELLER, HEATHER; Canadian Journal of Dietetic Practice & Research, 2016; 77(1): 3-8. 6p. (Article) ISSN: 1486-3847
Subjects: Questionnaires; Instrument Validation; Health Knowledge Evaluation; Attitude to Health Evaluation; Nutrition Education Evaluation; Middle Aged: 45-64 years; Aged: 65+ years; Male; Female
Academic Journal
PDF Full Text
Search NTU Catalog
Search NIE Catalog
2. Nutrition and Health Education Programme of ICDS Scheme for Pregnant Women in Rural Punjab.
Kular, Sarbjit Singh; International Journal of Nursing Education, Jan-Mar2016; 8(1): 139-144. 6p. (Article) ISSN: 0974-9349
Subjects: Nutrition Education In Pregnancy; Health Education In Pregnancy; Child Health Services India; Health Behavior; Health Resource

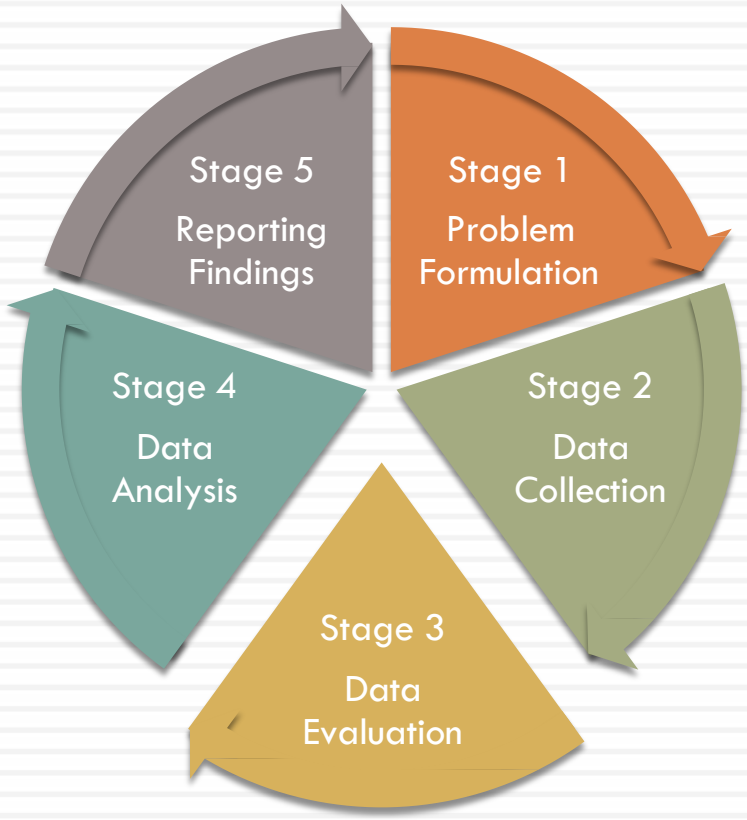

CNKI中国知网 www.cnki.net
總庫
檢索
出版來源
CNKI AI
我的CNKI
高級檢索
專業檢索
作者發文檢索
主題
精確
AND
作者
精確
AND
文獻來源
精確
OA出版
網絡首發
增強出版
中英文擴展
同義詞擴展
時間範圍:
發表時間
--
更新時間
不限
重置條件
檢 索

Concept Keywords:

- Artificial Intelligence: artificial intelligence, AI, machine learning, deep learning, natural language processing, NLP, neural network, intelligent algorithm, large language model, generative AI

- Health: health, healthcare, public health, health management, digital health, mobile health, mHealth, disease diagnosis, chronic disease management, health promotion, precision medicine

Stage 3: Data Evaluation





Data Coding

- Data extraction is the process by which researchers obtain the necessary information about study characteristics and findings
- Standardized data extraction provides consistency, thereby potentially reducing bias, improving validity and reliability
  - Ensures that the same information is obtained (extracted) from each study
- Should be developed early in the project
- Pilot the data extraction form with a sample of studies
  - Keep track of revisions, corrections, amendments
- If possible, pilot the exporting, analysis, and outputs of the data extraction form



Data Coding

- Purpose:
  - Provides an accounting of the research included in your review
  - Identifies key components of the research including interventions, subjects, and methods



Data Coding

- Levels of Coding
  - Abstract screening
  - Study eligibility screening
  - Study content coding

What Is Study Screening?

is the systematic process of **evaluating identified studies** against predetermined inclusion and exclusion criteria to **determine their eligibility** for a systematic review.

This process ensures that only relevant, high-quality studies that address the research question are included.

Key Note:

- It's how we pick which studies belong in our systematic review.
- Goal: keep only the studies that truly answer the research question.

Why It Matters

- Ensures methodological transparency and reproducibility
- Minimizes selection bias
- Maintains the validity and reliability of review findings
- Provides a clear audit trail of decision-making
- Helps manage potentially large volumes of literature efficiently

Two Screening Stages

- ❑Title & abstract screening (quick check).
- ❑Full-text screening (deep check).



Abstract Data Coding

- Abstract Screening
  - Is the study relevant based on the title and abstract
    - Apply title/ abstract eligibility criteria, which are often less specific than the full inclusion/exclusion criteria
      - If yes, retrieve full text
      - If unclear, retrieve full text
      - Exclude obviously irrelevant studies
    - Include two trained raters working independently if possible

Title and Abstract Screening (Level 1)

This initial screening phase involves:

Process:

- ❑ Reviewing titles and abstracts of all identified records
- ❑ Making rapid **inclusion/exclusion** decisions based on basic eligibility criteria
- ❑ Adopting an **"over-inclusive" approach**—when in doubt, include the study for full-text review

Key Considerations

- This stage typically eliminates 80-95% of identified records
- Speed is important, but accuracy is paramount
- Err on the side of inclusion to avoid missing relevant studies

Common Reasons for Exclusion

- Wrong population or setting
- Outcome not relevant.
- Irrelevant intervention or exposure
- Study design not meeting criteria
- Not addressing the outcome of interest
- Wrong publication type (e.g., editorials, letters)
- Not enough information!!!!!!!!!!!!!!!!!!!!!!!!

LEVEL 1 EXERCISE

Review question

Do smartphone applications improve glycemic control in adults with type 2 diabetes?

Inclusion criteria

- Adults aged 18 years or older with type 2 diabetes
- A smartphone application is the main component of the intervention
- Comparator: usual care, waiting list, or non-app intervention
- Randomized controlled trial
- Reports HbA1c measured at least 12 weeks after randomization
- Full-text journal article

Exclusion criteria

- Type 1 diabetes, gestational diabetes, or prediabetes
- Participants younger than 18 years
- Text-message-only, telephone-only, or website-only interventions
- Non-randomized studies
- Study protocols, conference abstracts, reviews, or editorials
- Does not report HbA1c
- Follow-up shorter than 12 weeks

①

Title

A smartphone-based self-management program for adults with type 2 diabetes: A randomized controlled trial

Abstract

A total of 240 adults with type 2 diabetes were randomly assigned to a smartphone application or usual care. The application provided blood glucose recording, dietary advice, medication reminders, and personalized feedback. HbA1c was measured at baseline, 12 weeks, and 24 weeks.

Decision

- Include/Obtain full text
- Exclude
- Unclear/Obtain full text

Reason: _____

②

Title

Text messaging to improve glycemic control in patients with type 2 diabetes

Abstract

A total of 180 adults with type 2 diabetes were randomly assigned to a text-messaging intervention or usual care. Participants received three messages per week about diet, exercise, and medication use. HbA1c was assessed after 24 weeks. No smartphone application was used.

Decision

- Include/Obtain full text
- Exclude
- Unclear/Obtain full text

Reason: _____

3

Title

A randomized evaluation of DiaCoach for diabetes management in primary care

Abstract

Patients were randomly assigned to the DiaCoach digital health program or usual care. The program helped participants record health information and provided self-management advice. Glycemic control was evaluated after 16 weeks. The abstract did not report the participants' age, type of diabetes, specific digital platform, or method used to measure glycemic control.

Decision

- Include/Obtain full text
- Exclude
- Unclear/Obtain full text
- **Reason: :** _____



Full-Text Data Coding

- Full-Text Eligibility Screening
  - Develop an eligibility screening form based on the inclusion / exclusion criteria
  - Complete for each study retrieved that is potentially eligible
    - Include reasons for ineligible studies (for PRISMA reporting)

Full-Text Screening (Level 2)

This more detailed screening phase involves:

- Obtaining and reviewing complete manuscripts
- Applying all eligibility criteria rigorously
- Documenting specific reasons for exclusion
- Identifying additional studies through reference lists (snowballing)

Key considerations

- Requires more time and careful evaluation
- Typically eliminates 50-70% of studies that passed title/abstract screening
- Decisions must be clearly documented with specific justifications

Challenges

- Accessing full texts (especially older or non-English publications)
- Interpreting insufficient methodological details
- Handling studies with mixed populations or outcomes

Best Practices in Study selection

- Dual Independent Screening:

The Gold Standard: Two reviewers independently screen each study at both levels.

Benefits:

- Reduces individual bias and error
- Increases reliability and validity
- Provides quality assurance

Process:

1. Both reviewers screen independently
2. Compare decisions and identify disagreements
3. Resolve conflicts through discussion or third-party arbitration
4. Calculate inter-rater reliability (Cohen's kappa)

Acceptable Kappa Values:

- $\kappa > 0.80$: Excellent agreement
- $\kappa = 0.60$-0.80: Good agreement
- $\kappa < 0.60$: May require recalibration

Managing Disagreements

When reviewers disagree:

Primary Strategy: Discussion between the two reviewers to reach consensus

Secondary Strategy: Involvement of a third reviewer to arbitrate

Documentation: Record all disagreements and resolution methods

Handling Special Cases

1. Multiple Publications from Same Study:

Link related publications

Treat as single study but extract data from all sources

Use most complete or recent data

Handling Special Cases

2. **Studies with Incomplete Information**:

Attempt to contact authors for clarification

Set reasonable waiting periods (typically 2-4 weeks)

Document all contact attempts

Make decisions based on available information if no response

Language Barriers:

- Use translation services or multilingual team members
- Consider language restrictions carefully (may introduce bias)
- Document translation methods

LEVEL 2 EXERCISE

Review question

Do smartphone applications improve glycemic control in adults with type 2 diabetes?

Inclusion criteria

- Adults aged 18 years or older with type 2 diabetes
- A smartphone application is the main component of the intervention
- Comparator: usual care, waiting list, or non-app intervention
- Randomized controlled trial
- Reports HbA1c measured at least 12 weeks after randomization
- Full-text journal article

Exclusion criteria

- Type 1 diabetes, gestational diabetes, or prediabetes
- Participants younger than 18 years
- Text-message-only, telephone-only, or website-only interventions
- Non-randomized studies
- Study protocols, conference abstracts, reviews, or editorials
- Does not report HbA1c
- Follow-up shorter than 12 weeks

①

Full-Text Article A

- Design: Parallel-group randomized controlled trial
- Participants: 240 adults with type 2 diabetes
- Intervention: A smartphone application providing blood glucose recording, dietary advice, medication reminders, and personalized feedback
- Comparator: Usual care
- Follow-up: 24 weeks
- Outcomes: HbA1c, body weight, medication adherence, and quality of life
- Publication type: Full-text journal article

Decision

- Include
- Exclude

Reason: _____

②

Full-Text Article B

- Design: Randomized controlled trial
- Participants: 150 adults with type 2 diabetes
- Intervention: Weekly video consultations with a nurse using a smartphone video-calling application
- The application did not provide blood glucose recording, education, reminders, or personalized feedback
- Comparator: Usual care
- Follow-up: 24 weeks
- Outcome: HbA1c
- Publication type: Full-text journal article

Decision

- Include
- Exclude

Reason: _____

3

Full-Text Article C

- Design: Cluster-randomized controlled trial
- Participants: 520 adults with diabetes, including 410 with type 2 diabetes and 110 with type 1 diabetes
- Intervention: A smartphone diabetes self-management application
- Comparator: Usual care
- Follow-up: Six months
- Outcome: HbA1c
- Results were reported for all participants combined
- Separate results for participants with type 2 diabetes were unavailable

Decision

- Include
- Exclude

Reason: _____

Tools & Technology You Can Use

- Covidence: User-friendly, widely used in health research
- DistillerSR: Highly customizable, robust features
- Rayyan: Free, AI-assisted screening
- EPPI-Reviewer: Advanced features for complex reviews
- RevMan: Cochrane's review management tool
- EndNote / Zotero
- Excel templates

Emerging Technologies

Machine Learning and AI:

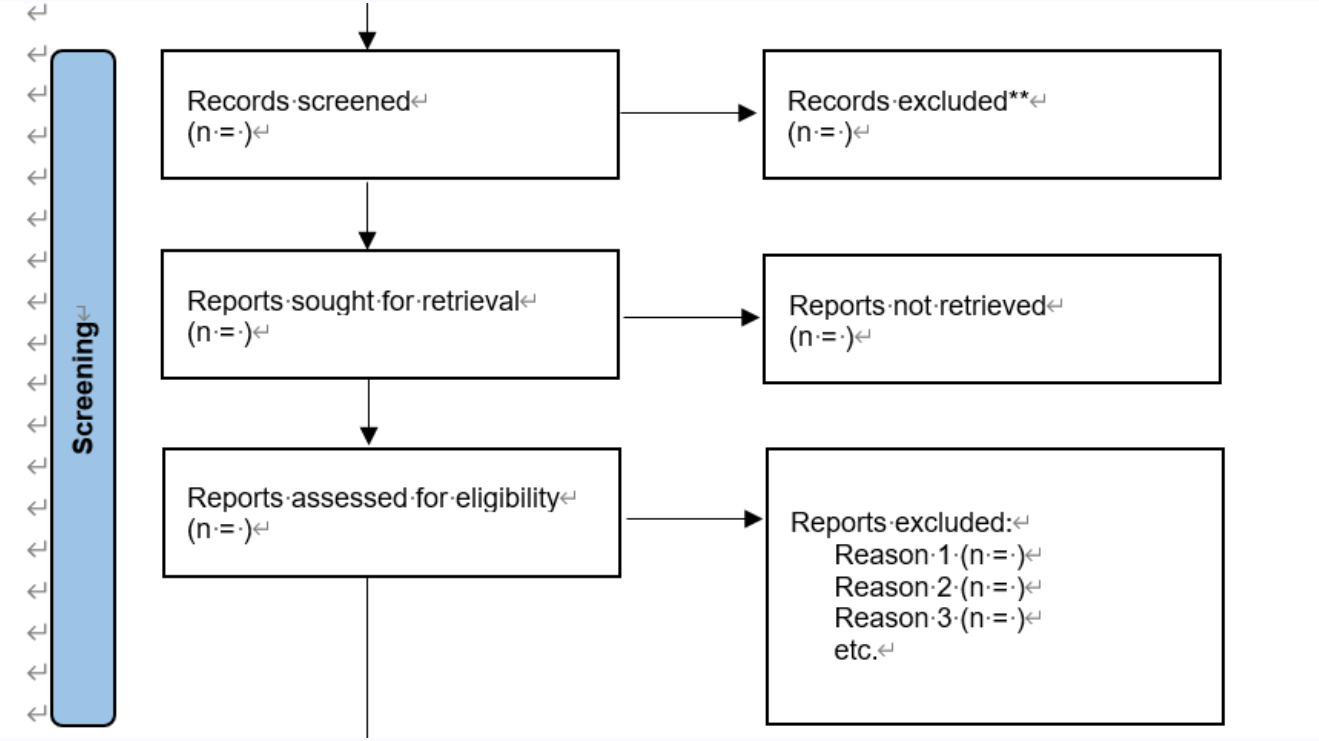
- Can prioritize studies likely to be relevant
- Reduces screening workload
- Still requires human oversight and validation
- Most effective after initial training set

Text Mining:

- Automates initial sorting based on keywords
- Identifies patterns in abstracts
- Supports decision-making but doesn't replace it

PRISMA Diagram

- Shows how many studies were found, screened, and kept.
- Used in all systematic review papers.



Key Takeaways

- Apply criteria the same way every time.
- Be transparent about decisions.
- Document everything clearly.



Study Content Coding

Data extraction form:

- Be consistent in the order and style you use to describe the information from each study
- Provide instructions, decision rules, and training on the use of the data extraction form
- Practice using the data extraction form
  - Compare completed pilot forms
  - Ensure that there is consistency in extracting data
  - Ideally two researchers should independently perform data extraction
- Record any missing information as unclear or not described
  - Makes it clear that the information was not included in the study rather than you forgot to extract it

Study Content Coding

Box 1.4 Example information requirements for data extraction

General information

Researcher performing data extraction

Date of data extraction

Identification features of the study:
- Record number (to uniquely identify study)
- Author
- Article title
- Citation
- Type of publication (e.g. journal article, conference abstract)
- Country of origin
- Source of funding

Study characteristics

Aim/objectives of the study

Study design

Study inclusion and exclusion criteria

Recruitment procedures used (e.g. details of randomisation, blinding)

Unit of allocation (e.g. participant, GP practice, etc.)

Participant characteristics

Characteristics of participants at the beginning of the study e.g.
- Age
- Gender
- Ethnicity
- Socio-economic status
- Disease characteristics
- Co-morbidities

Number of participants in each characteristic category for intervention and control group(s) or mean/median characteristic values (record whether it is the number eligible, enrolled, or randomised that is reported in the study)

(Continued)

Intervention and setting

Setting in which the intervention is delivered

Description of the intervention(s) and control(s) (e.g. dose, route of administration, number of cycles, duration of cycle, care provider, how the intervention was developed, theoretical basis (where relevant))

Description of co-interventions

Outcome data/results

Unit of assessment/analysis

Statistical techniques used

For each pre-specified outcome:
- Whether reported
- Definition used in study
- Measurement tool or method used
- Unit of measurement (if appropriate)
- Length of follow-up, number and/or times of follow-up measurements

For all intervention group(s) and control group(s):
- Number of participants enrolled
- Number of participants included in analysis
- Number of withdrawals, exclusions, lost to follow-up
- Summary outcome data e.g.
  - Dichotomous: number of events, number of participants
  - Continuous: mean and standard deviation

Type of analysis used in study (e.g. intention to treat, per protocol)

Results of study analysis e.g.
- Dichotomous: odds ratio, risk ratio and confidence intervals, p-value
- Continuous: mean difference, confidence intervals

If subgroup analysis is planned the above information on outcome data or results will need to be extracted for each patient subgroup

Additional outcomes

Record details of any additional relevant outcomes reported

Costs

Resource use

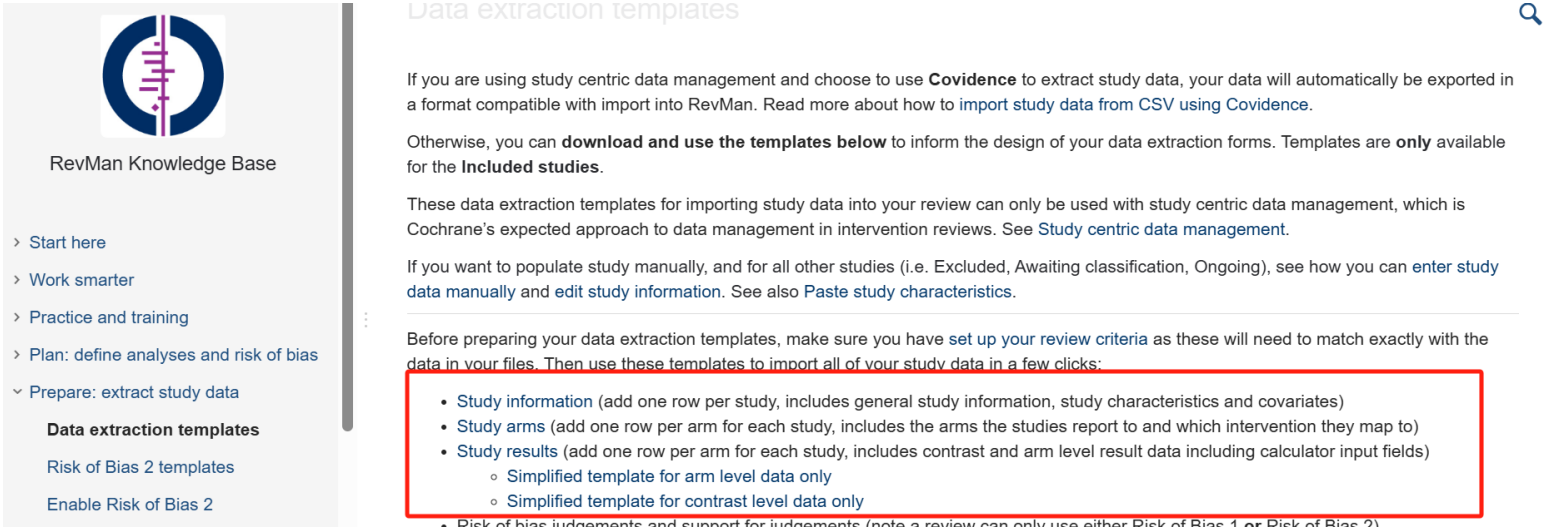
Adverse events

NB: Notes fields can be useful for occasional pieces of additional information or important comments that do not easily fit into the format of other fields.

Systemtaic reviews: CRD's guidance for undertaking systematic reviews in health care. 2009. Retrieved from: https://www.york.ac.uk/inst/crd/index_guidance.htm

Study Content Coding

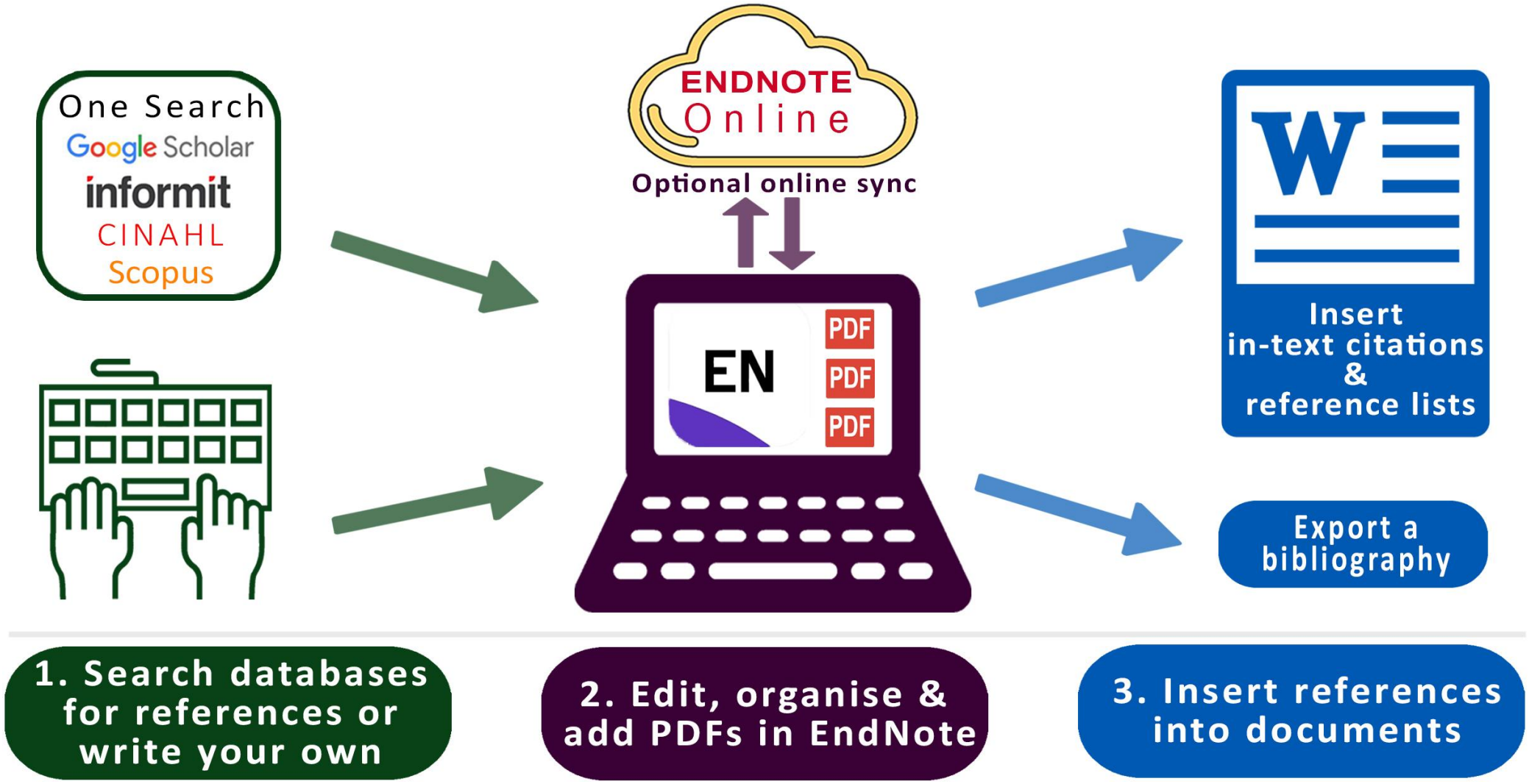
- Cochrane has developed templates that can be revised
  - Google: “data extraction form” cochrane
- Review published systematic reviews relevant to your research area
  - Request permission to adapt the data extraction form





Study Content Coding

- Software for data extraction
  - RevMan (Cochrane – can use free of charge for academic purposes)
  - EPPI-Reviewer (one month free trial)
  - JBI Sumari (subscription based)
  - NVivo (UofC site license)
  - Abstraktr
  - Excel


One Search
Google Scholar
informit
CINAHL
Scopus
ENDNOTE
Online
Optional online sync
EN
PDF
PDF
PDF
Insert
in-text citations
&
reference lists
Export a
bibliography
1. Search databases for references or write your own
2. Edit, organise & add PDFs in EndNote
3. Insert references into documents

Contents

Systematic Review context

Managing downloads

De-duplication

Screening

Find full text

Backups

Sharing whole libraries or groups with collaborators

EndNote helps with…

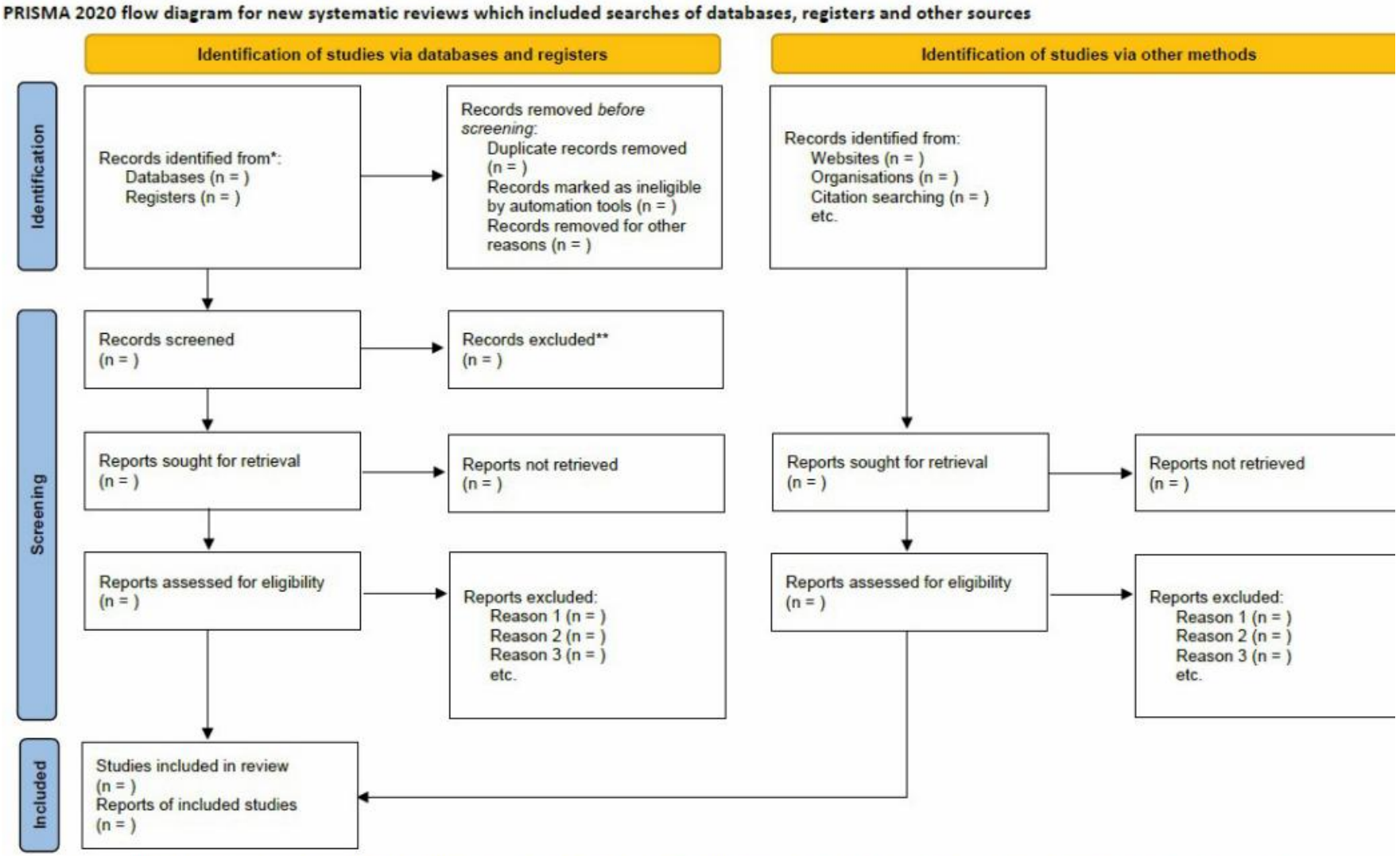
Managing downloads

(De-duplication)

(Screening)

Find full text

Citations in write-up



*Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers).
**If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools.

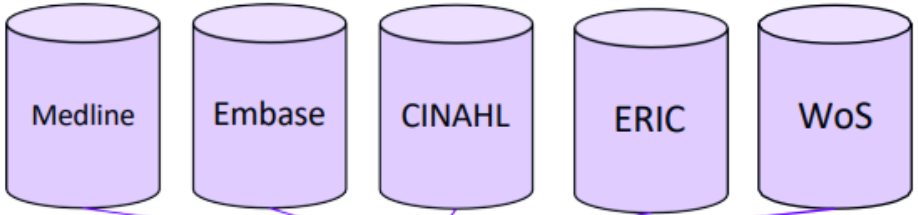
From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372: doi: 10.1136/bmj.n71. For more information, visit: http://www.prisma-statement.org/

Workflow

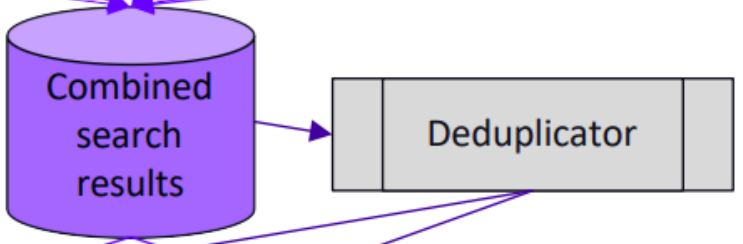
(Each container represents a separate EndNote library)

This workflow ensures you have a backup and record for search results at each stage

Create and keep an EndNote Library for each batch of search results

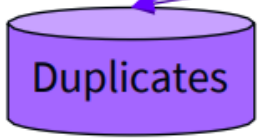


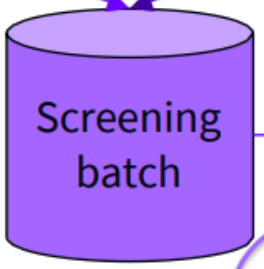
Copy and paste results into a combined EndNote library for all databases

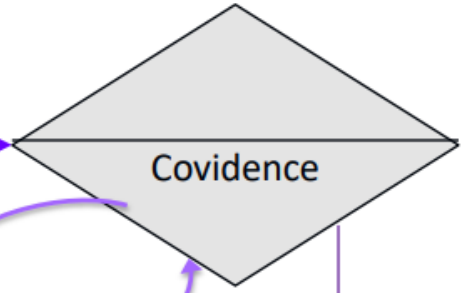


Run a de-duplication, either within EndNote or an external tool like the SR Accelerator's Deduplicator tool

Don't delete duplicates – keep them set aside either in a group or separate EndNote Library

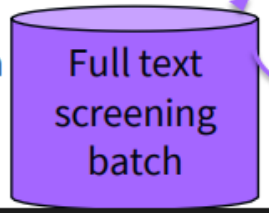






Screen results either in EndNote or using an external tool like Covidence

EndNote can quicken getting the full text journal articles





Use EndNote for inserting references into final write-up

Managing downloads

- Bulk downloads and keeping track of results numbers
- A cautious approach that provides backups
  - separate libraries for each results set
  - combined library of all results (back-up)
  - working copy of combined library for deduplication and screening

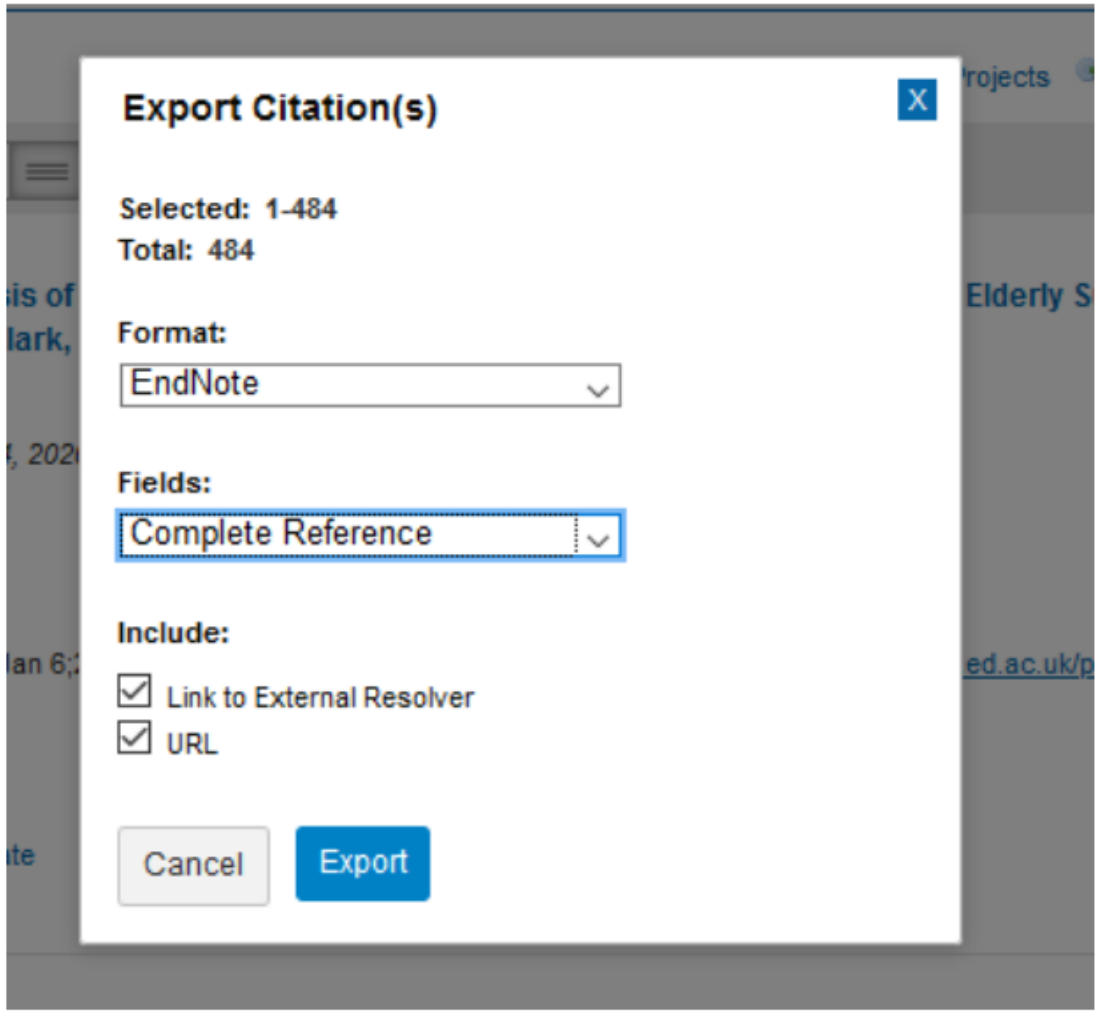
Bulk downloads

Screening against criteria for inclusion is best done offline

Use bulk downloads into EndNote

Be sure to note results numbers and check against import numbers

Be sure to select download options that include e.g. abstracts and DOIs, to make screening phases easier



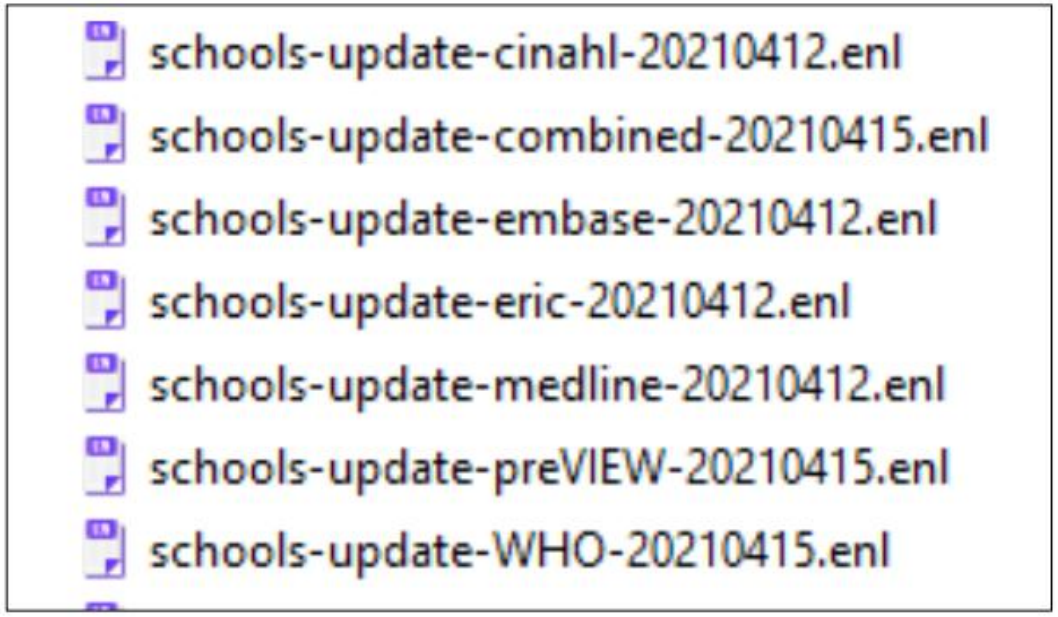
Libraries for each results set

This project example had six database results sets

One combined results library, from which duplicate results were copied in to separate library

Allows each set of results to be checked later if needed

Allows easy rescue if combined library is lost/corrupted



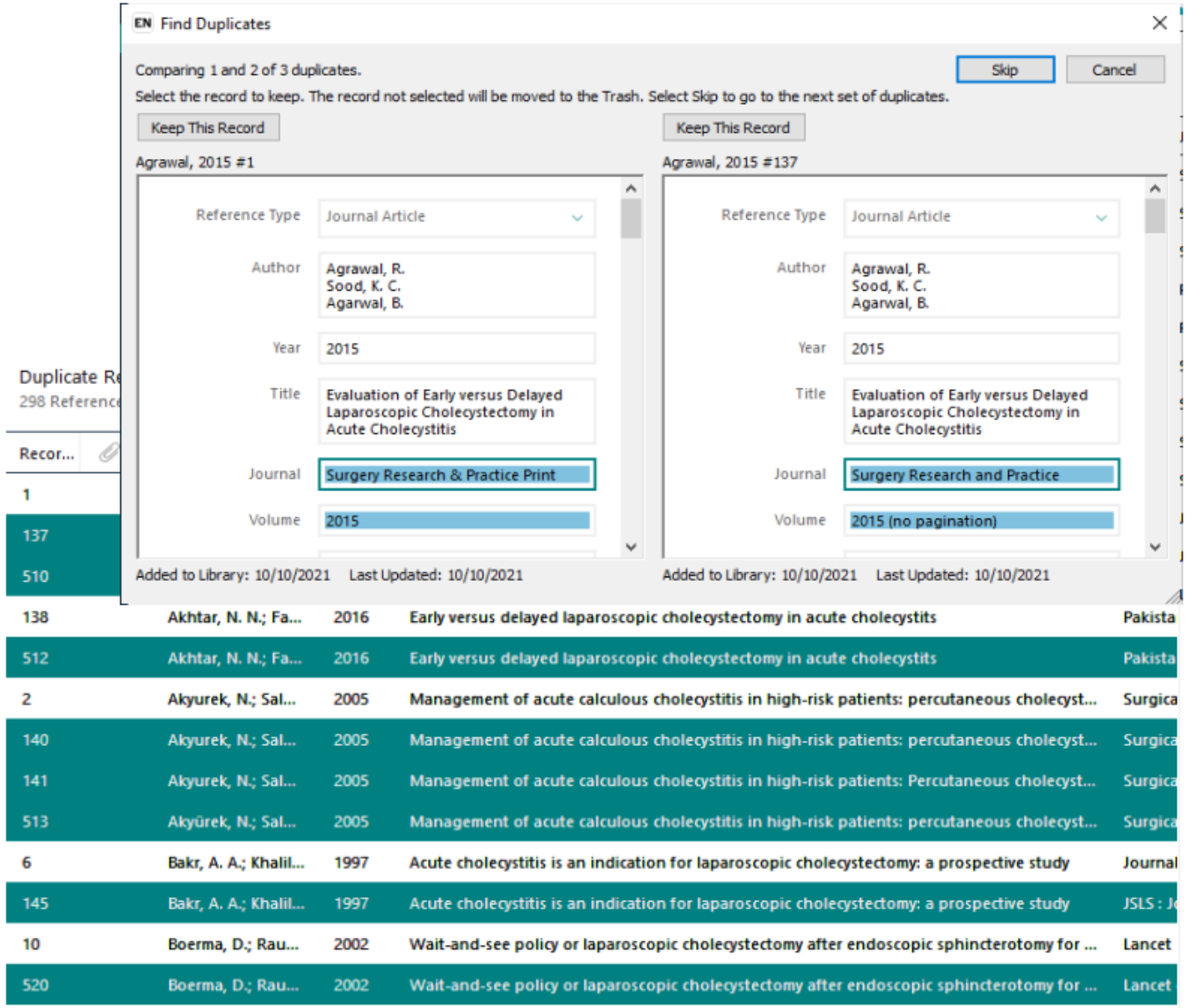
De-duplication 1

EndNote can be used for deduplication prior to screening

If you have a lot of duplicates, close the comparison window and screen list, to save time

Beware! The duplicates list shows both first copy (not highlighted) and duplicate copies (highlighted), so take care only to move duplicates.

Note: EndNote will miss duplicates with the default criteria. Consider adjusting criteria, but check carefully items identified as duplicate

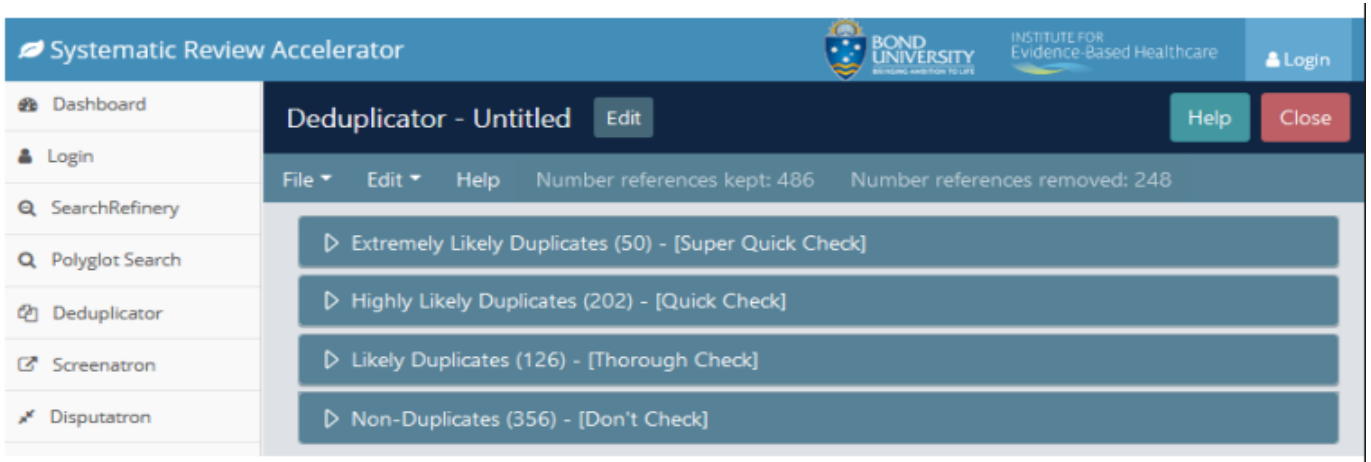


De-duplication 2

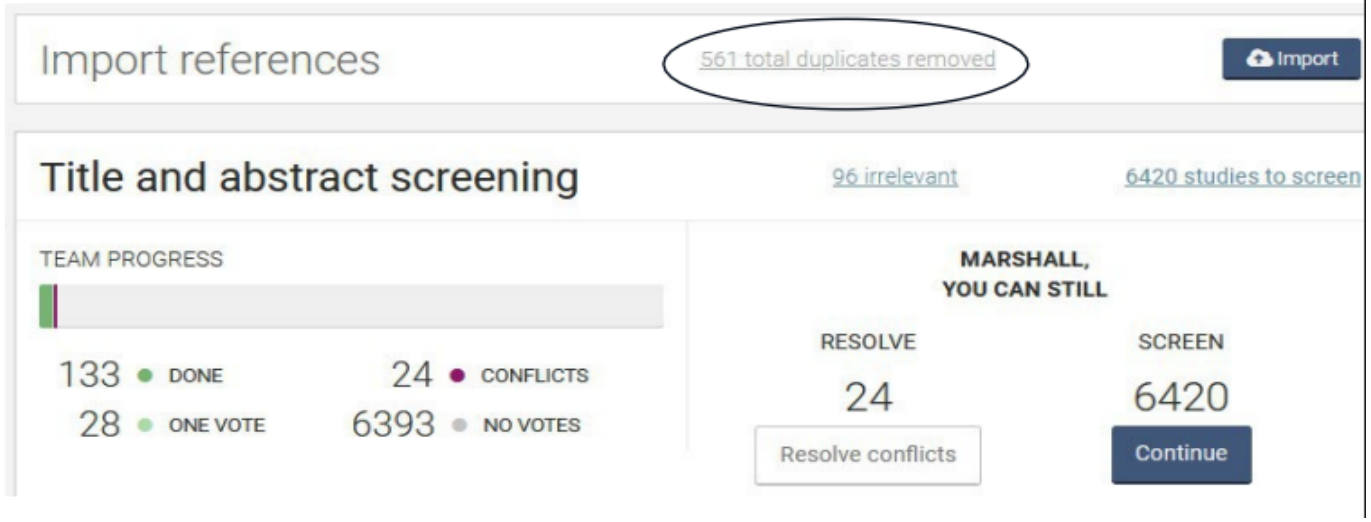
External tools:

Deduplicator available from the Systematic Review Accelerator

https://sr-accelerator.com/#/deduplicator



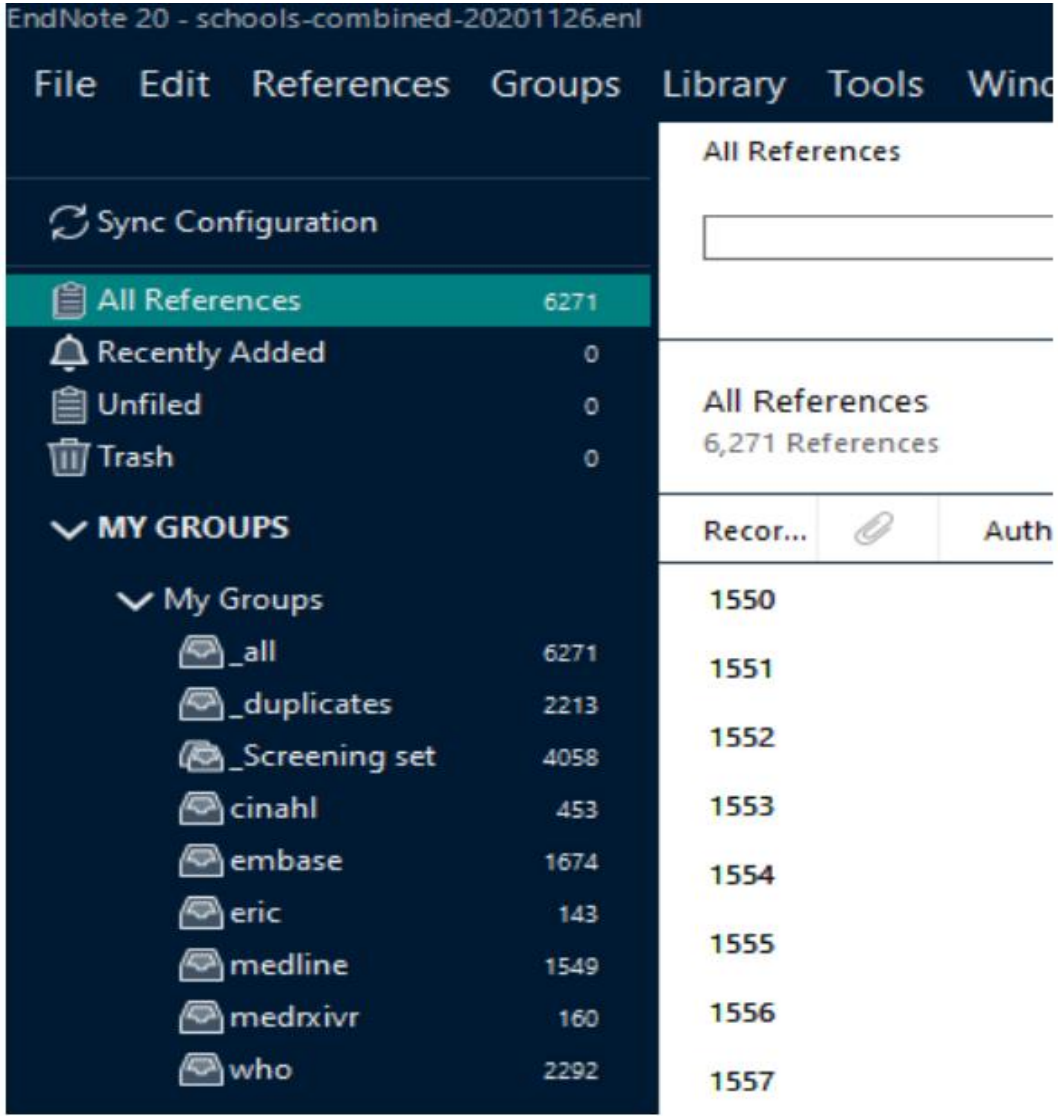
Covidence, which you may want to use for screening, also checks for duplicates as part of upload process.



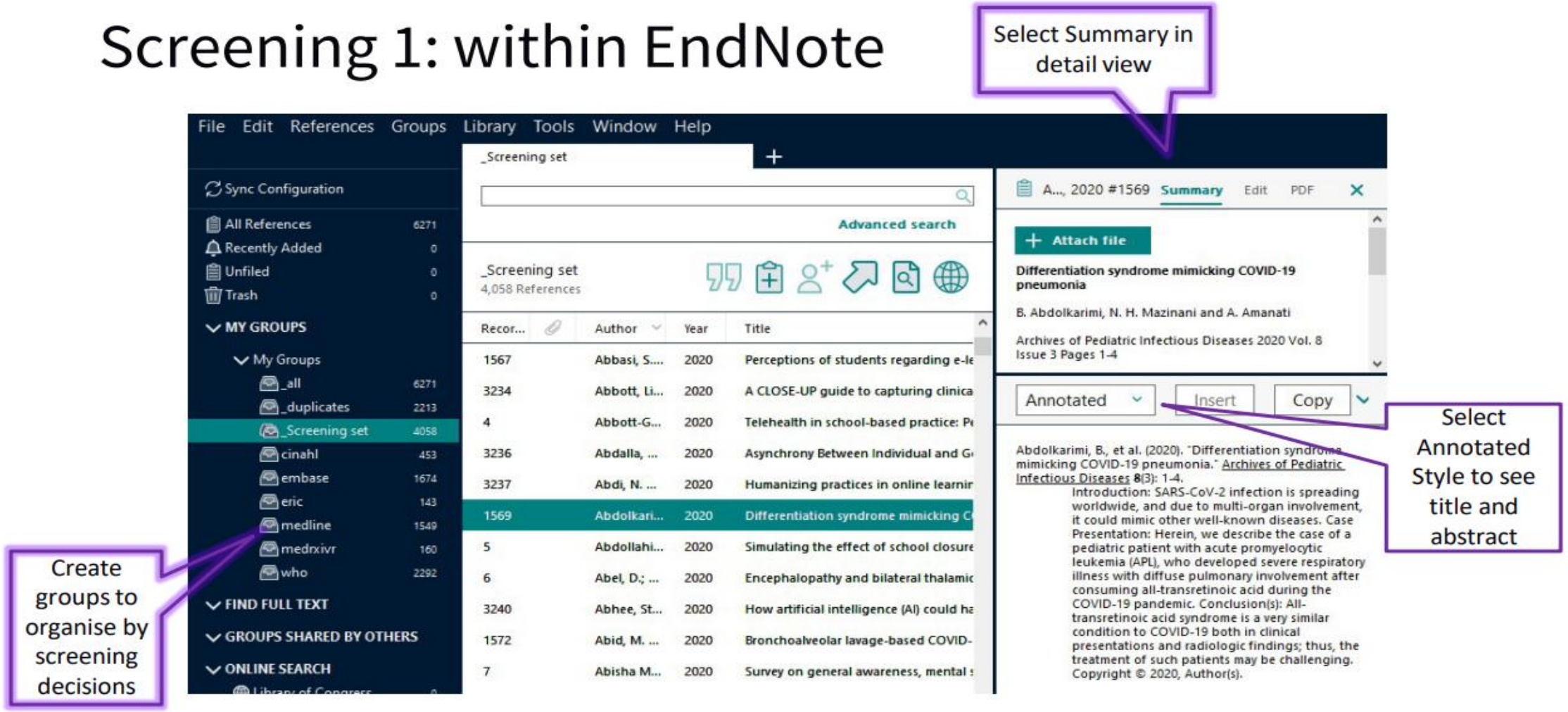
Using Groups

EndNote groups can be used to organise the records

- Keep track of screening decisions
- Keep track of database/origin
- Plan workload in team

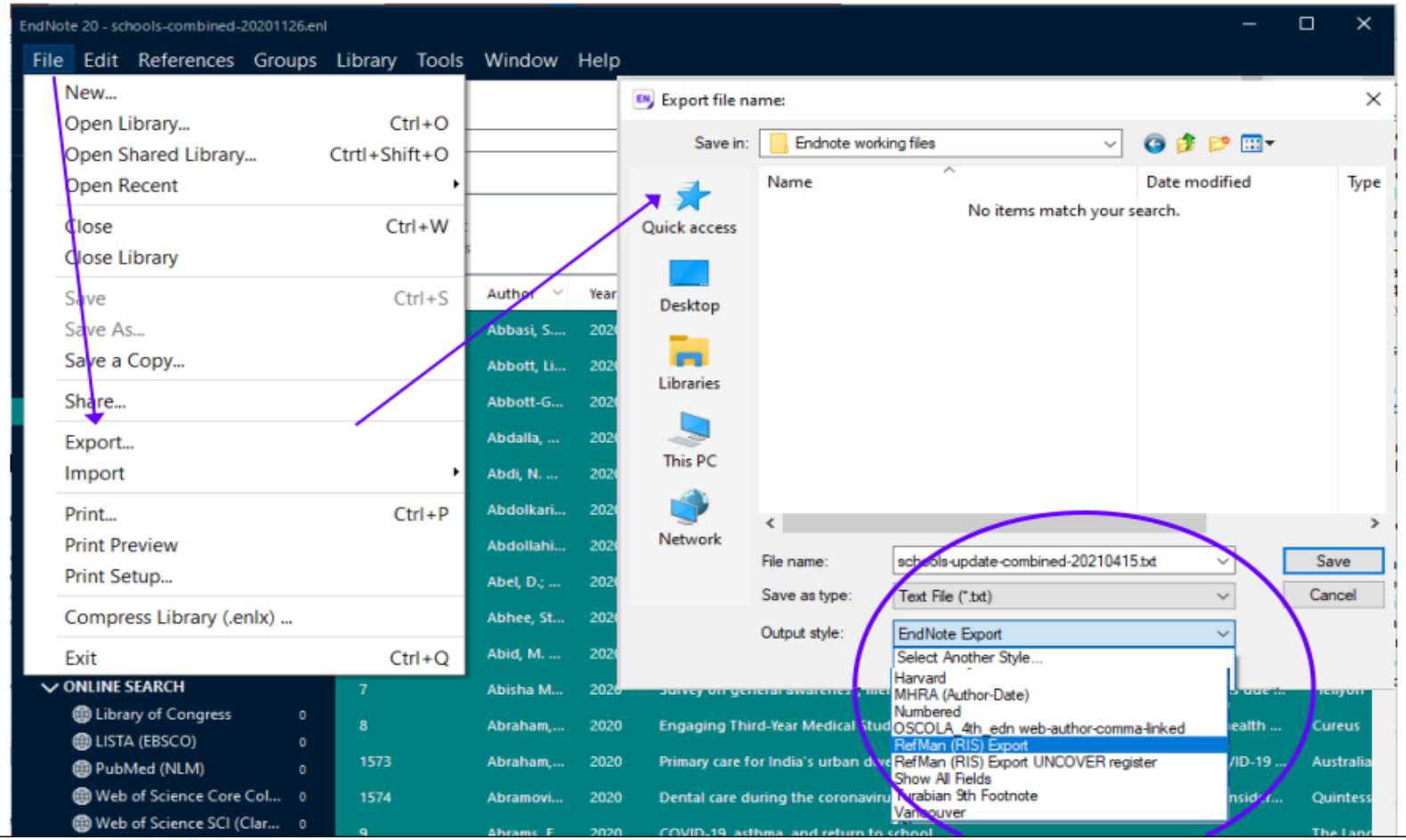


Screening 1: within EndNote



Screening 2: exporting in RIS or xml

Reference Manager (RIS) exported as text file type .txt is a standard format, but some tools (e.g. Covidence) may also work with EndNote export as file type .xml



Screening 3: Exporting to excel

For export to Excel, use the style, BasicExportToExcel

- Select all relevant records
- Right-click and Copy Formatted
- Paste into Excel using right-click menu
- Internal line breaks within the record text cause problems



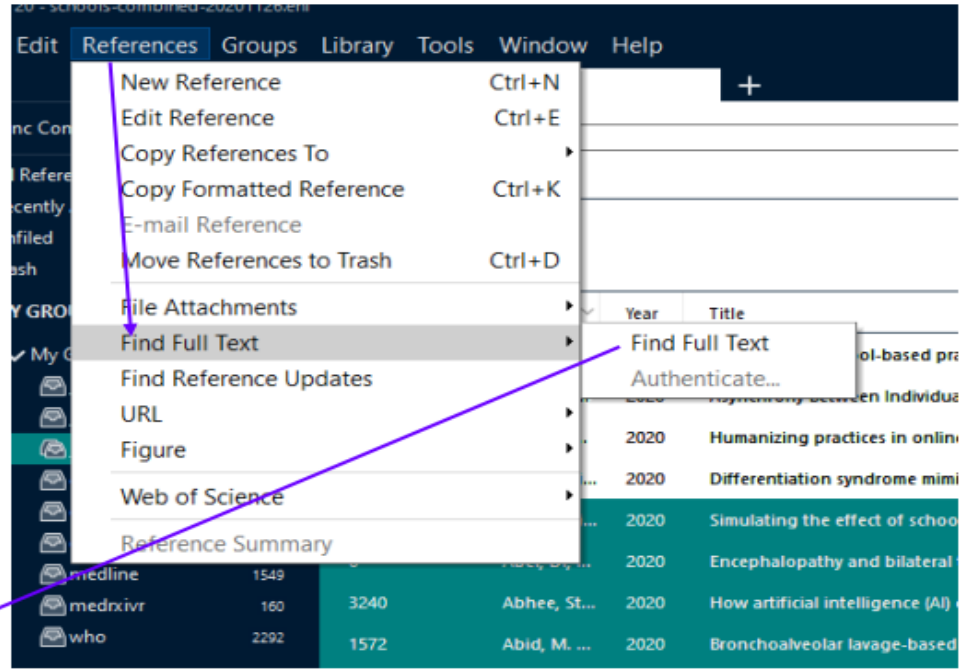
D	E	F	G	H	I	P	
Referei	Author	Year	e pub date	Title	Secondary Title	DOI	Abstract
Journal Ar	M. C. Alza	2020	19/04/2020	Severe COVID-19 during Pregnancy and Possible Vertical Transmission	Am J Perinatol	10.1055/s-	There are f (COVID-19) There is lin

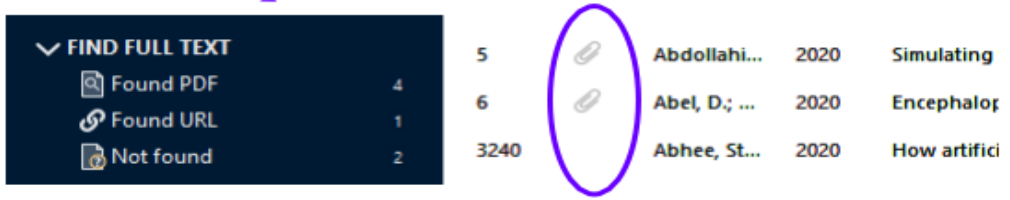
Find full text

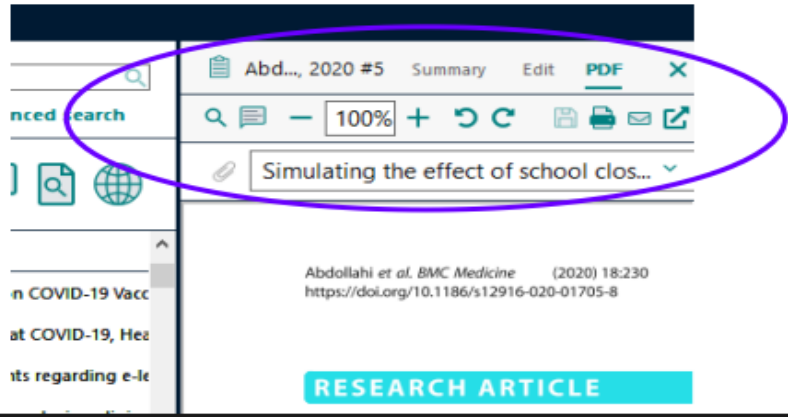
Save time!

Use UoE VPN for best results

Works best for journal articles with DOIs

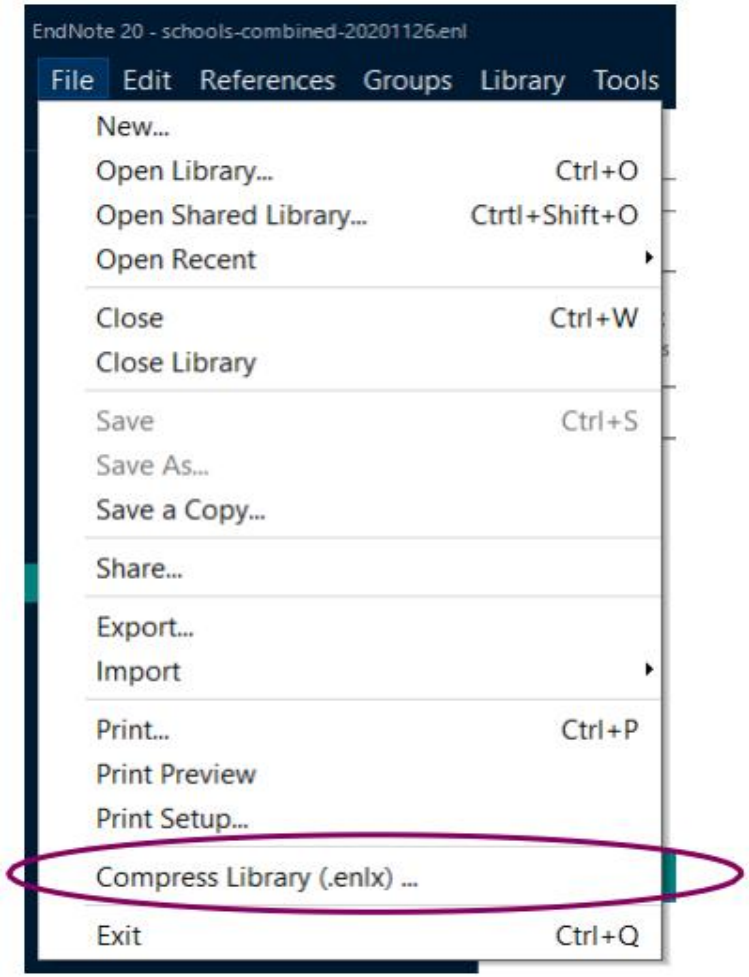






Backups

Create a compressed library at key stages and save somewhere that is backed up – e.g. OneDrive



Sharing with collaborators



Beware of using cloud servers for storing Libraries

Sharing goes via EndNote Online

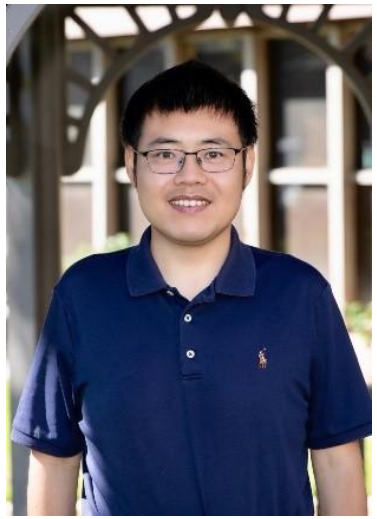
Can share

- Use Groups in EndNote Online
- whole Libraries https://support.clarivate.com/Endnote/s/article/EndNote-Library-sharing-and-editing?language=en_US

Alternative: Compressed Library on shared fileserver – nominate one team member to keep file version up to date


Thank you!

Team for AI-Empowered Smart Healthy Ageing





Visiting Student Recruitment

For Juniors/Seniors & 1st-Year Graduates







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